

Structured Extraction of PICO Information from Clinical Trial Abstracts: A Comparative Study of Rule-Based, Prompted and Fine-Tuned Approaches

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Group 2 Introduction to AI and Text Analytics Coursework

Github link: <https://github.com/Blacknomizi/IATA-Coursework-Group2>

Abstract

Converting clinical trial abstracts to structured tables based on the PICO format has been one of the most important ways to enable large-scale evidence synthesis. Using the EBM-NLP corpus (Nye et al., 2018), we compare nine extraction systems along two design axes namely rule-based versus LLM-based, and end-to-end versus decomposed, on a locked 50-document evaluation set, and report token-overlap F1, strict span F1, coverage and downstream keyword query usability. Fine-tuned transformer models clearly dominate: end-to-end BioBERT achieves a macro token-overlap F1 of 0.673, narrowly ahead of fine-tuned DistilBERT (0.647) and well above any prompted LLM (best 0.347 for zero-shot flan-T5). Three findings emerge as more interesting than the headline ranking. Firstly, zero-shot beats few-shot for prompted flan-T5 because few-shot instances leak study-specific phrases (*Rhinocort Study Group*) as learned priors. Secondly, the value of decomposing extraction depends on the underlying model: it helps weak QA-based extractors (+0.12 macro F1) but hurts strong supervised ones (−0.07 strict span F1) because the relevance filter over-discards. Third, high coverage does not imply usability, prompted LLMs return some text for 100% of documents but retrieve only 1 of 11 placebo-controlled trials in downstream queries, while every fine-tuned and rule-based system finds ≥ 11 .

1 Introduction

Systematic reviews and meta analyses underpin evidence-based medicine, but a practising clinician evaluating a treatment must screen thousands of trial abstracts before identifying a few dozen suitable studies. The PICO schema: *Population, Intervention, Comparator, Outcome* is the standard way of describing a trial in structured form for retrieval and synthesis (Higgins et al., 2019). Automated PICO extraction would convert free text abstracts into a queryable table, allowing reviewers to filter

trials by population, group trials with shared interventions or compute effect sizes across studies that report comparable outcomes.

We work with EBM-NLP (Nye et al., 2018), a corpus of 4,993 abstracts annotated for participants (P), interventions (I) and outcomes (O) at token level. The dataset combines crowd-worker annotations (*starting_spans*, used in the rule-based pipeline of 02_rule_based_vs_llm.ipynb) with expert-aggregated hierarchical labels (*aggregated/hierarchical_labels*, used by every supervised model). Comparators are folded into the Interventions field by design (Nye et al., 2018, §2); we extract only P, I, O, and discuss the comparator (the “C” in PICO) as a dataset level limitation in §4.

Requirements for a good system. A reviewer using the extracted table cares about three things, in order of decreasing severity. Recall matters most because a missed trial is silently dropped from a meta-analysis and cannot be recovered downstream. Coverage matters next because a row with empty PICO cells is unusable regardless of the precision of its populated fields. Precision matters last because false positives can be filtered manually but false negatives cannot. F1 alone hides the recall-coverage trade-off, so we report all three throughout this paper.

Example. A typical abstract from the corpus illustrates what extraction must recover. PMID 10070173 begins: “*Comparison of budesonide Turbuhaler with budesonide aqua in the treatment of seasonal allergic rhinitis. Two hundred and eighty-four out-patients...*”. The gold annotations mark seasonal allergic rhinitis and Two hundred and eighty-four out-patients as Participants, budesonide Turbuhaler and budesonide aqua as Interventions, and quality of life and Mean daily nasal symptom scores (later in the abstract) as Outcomes. A correct system must there-

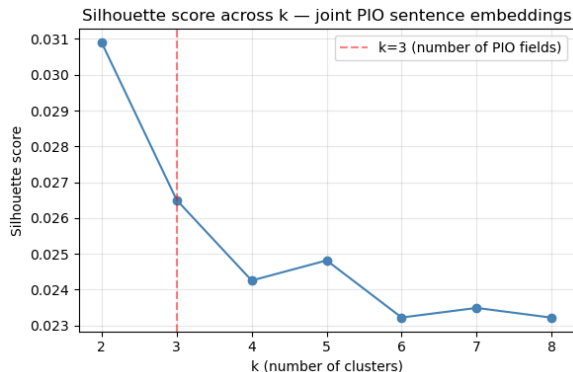


Figure 1: Silhouette score across k for k -means on joint-PIO sentence embeddings (MiniLM, 3,000 sentences). Silhouette declines monotonically from $k=2$ to $k=8$ with no meaningful peak at $k=3$, and ARI and NMI against gold sentence-level labels are 0.012 and 0.007 essentially random. This null result motivates supervised extraction (§2).

fore handle multi-span, multi-sentence extraction with both numerical and noun-phrase entities; cohort sizes and condition descriptors both qualify as Participants under the EBM-NLP scheme.

Data exploration Before designing extractors we asked whether sentence-level structure already separates the schema fields. We embedded all training sentences with MiniLM (Reimers and Gurevych, 2019), clustered them with k -means and HAC at $k = 2 \dots 8$, and computed adjusted Rand index, normalised mutual information and silhouette against gold sentence-level labels (Figure 1). The result is a clear negative: ARI = 0.012 and NMI = 0.007 against gold for joint PIO at $k = 3$ essentially indistinguishable from random and silhouette is roughly flat across all k . Sentence embeddings do not naturally recover the schema, which motivates supervised extraction (§2) rather than an unsupervised pipeline. A biomedical encoder (PubMedBERT-MS-MARCO (Deka and Pal, 2022)) yields a small but consistent +0.018 ARI on the same data, supporting the use of domain pre-trained encoders for downstream extraction without changing the headline conclusion.

Contributions We have compare nine extraction systems along two explicit design axes (rule-based vs LLM; end-to-end vs decomposed) on an identical 50 document locked evaluation set; we share data loading, prediction storage and evaluation through (src/) so all numbers in this report are computed under the same metric, with the same gold spans, on the same documents. *Note: NBO**

in the entire report refers to respective ipynb file in notebook folder on github

2 Methods

2.1 Shared infrastructure and preprocessing

Every system in this study reads from the same loader, writes predictions to a JSON file in predictions/, and is graded by the same evaluator on the same 50 documents. The locked evaluation set is the first fifty document IDs in the intersection $\text{TEST}_P \cap \text{TEST}_I \cap \text{TEST}_O$, sorted, and saved to eval_doc_ids.json; reusing this list across notebooks guarantees comparability when individual notebooks are re-run in isolation. Hierarchical EBM-NLP labels (0–7) are flattened to BIO tags by hierarchical_to_bio, while the rule-based pipeline uses the simpler starting_spans layer because regex patterns operate on token sequences rather than typed entities.

BIO tag scheme. For each PICO field $t \in \{P, I, O\}$ independently, a token sequence $\mathbf{x} = (x_1, \dots, x_n)$ receives binary BIO tags $y_i \in \{B, I, O\}$. Hierarchical EBM-NLP labels ($r_i \in \{0, \dots, 7\}$) collapse to BIO via:

$$y_i = \begin{cases} O & \text{if } r_i = 0 \\ B & \text{if } r_i \neq 0 \text{ and } r_{i-1} = 0 \\ I & \text{if } r_i \neq 0 \text{ and } r_{i-1} \neq 0 \end{cases} \quad (1)$$

The per field decomposed pipelines (NB04 DistilBERT, NB05 QA, NB06 decomposed BioBERT) train one binary tagger per field. The end-to-end BioBERT (NB06) instead uses a merged 7-tag schema ($O, B-P, I-P, B-I, I-I, B-O, I-O$) so a single model emits all three field assignments jointly. (Here all the NB0* refers to respective notebooks (ipynb) from github repo in notebook folder)

We adopt two evaluation metrics. Token-overlap F1 (lenient, bag-of-tokens) is used to compare all nine approaches, because not all of them emit BIO sequences. Strict span F1 (exact start/end/type match) is used inside the supervised notebook to compare end-to-end versus decomposed under a stricter rule (§3.1). Coverage is the fraction of documents with at least one non-empty extraction; downstream usability is measured by four keyword queries against the table (§3).

2.2 Pipeline architectures

Figure 2 shows the four pipeline architectures implemented in this study. All 4 read the same EBM-

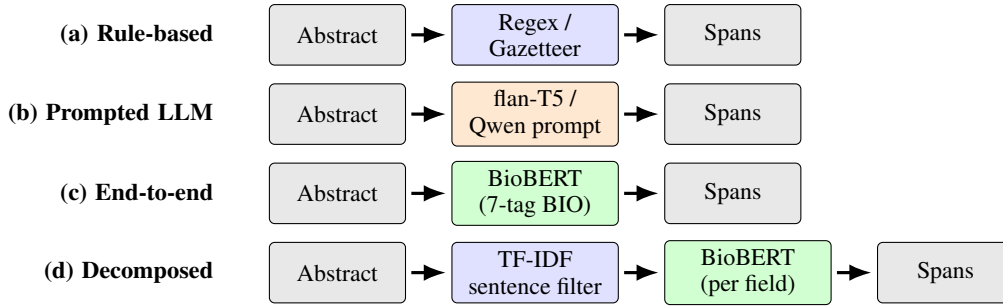


Figure 2: The four pipeline architectures compared are sharing common input (*Abstract*) and output (*Spans*) but differing in the intermediate processing. (a) Rule-based extractors apply hand-crafted patterns or a frequency gazetteer directly to the tokenised abstract. (b) Prompted LLMs wrap the abstract in a question prompt and parse generated text. (c) End-to-end transformers (BioBERT) tag every token with a single 7-tag BIO label. (d) Decomposed transformers first filter sentences with a TF-IDF classifier, then run a per-field BioBERT tagger only on accepted sentences.

NLP abstract tokens and emit a per field span list, but differ in how they map between the two: rule-based pipelines uses deterministic pattern matching, prompted LLMs treat extraction as text generation, end-to-end transformers tag every token with a joint BIO label, and decomposed transformers split the work into a sentence-relevance filter and a per-field tagger.

2.3 Design axis A: rule-based vs LLM

The first axis asks how much of the task can be solved by symbolic patterns over the abstract, and how much requires a learned model. We implement four extractors that span this axis from purely hand-crafted to fully learned. The simplest is a regex-based extractor (NB02) using hand-written patterns for cohort sizes (`\d+\s+(patients|men|women)`), drug doses (`\d+\s*mg`), and outcome cues such as side effects and score. One step up in data-drivenness is the gazetteer (NB02), which extracts the top-200 1- to 3-grams per field by frequency in 300 training-set gold spans and matches them at test time as whole-word case-insensitive regex; this asks how far one can get by simply memorising training-time spans. We then introduce a small fine-tuned LLM, Qwen3-0.6B (NB02), an instruction-tuned decoder (Qwen Team, 2024) given three in-context examples per field and asked to copy the answer. Finally, three prompt variants on flan-T5-base (250M parameters, Chung et al., 2022) (NB03) let us isolate the effect of the prompt itself: zero-shot question-answering, three-shot worked examples, and three-shot with a JSON output constraint. All flan-T5 generations use beam search ($n = 4$), 512-token input, and 80–120 output tokens.

2.4 Design axis B: end-to-end vs decomposed

The second axis asks whether a single model should emit all three field assignments at once, or whether the task should be broken into *find the relevant sentence*, then *extract the span*. Five systems span this axis. The CRF baseline (NB06) is a linear-chain CRF (Lafferty et al., 2001) with shape, prefix/suffix and medical-keyword features; we train one tagger per field and merge predictions with a $P < I < O$ priority rule. It is the classical baseline against which deep models are usually judged. Fine-tuned DistilBERT (04_transformer_finetuned.ipynb) takes the same per-field decomposition into the deep-learning regime: three independent DistilBERT-base-uncased (Sanh et al., 2019) token classifiers (one per field) trained on hierarchical NER tags (B-1 . . . I-4 per field) for 5 epochs at batch 16, learning rate 1×10^{-5} and weight decay 0.01.

The end-to-end BioBERT (06_end_to_end_vs_decomposed.ipynb) does the opposite: a single BioBERT-base-cased (Lee et al., 2020) fine-tuned on a merged 7-tag schema (O, B-P/I/O, I-P/I/O), with two-phase training that first trains the classification head for 4 epochs with the encoder frozen, then fine-tunes the full model for 2 epochs at 2×10^{-5} . The decomposed BioBERT (06_end_to_end_vs_decomposed.ipynb) tests whether the per-field decomposition of 04_transformer_finetuned.ipynb also helps when the underlying model has already seen biomedical text: a TF-IDF + logistic-regression classifier first predicts whether each sentence contains an entity of a given type, then a per-field BioBERT token tagger runs only on accepted sentences. Finally, extractive QA (NB05)

uses deepset/roberta-base-squad2 (Liu et al., 2019) in three configurations end-to-end (one question per field on the whole abstract), 2-step decomposed (sentence classification by argmax confidence, then extraction), and 3-step decomposed (2-step plus post-hoc cleaning: deduplication, ≥ 2 -token spans, and a “person word” filter for participants).

2.5 Hypotheses

We motivate the comparison with five testable claims. **H1:** LLM-based extractors will beat rule-based extractors, because the latter cannot generalise beyond the patterns the author thought of. **H2:** Few-shot prompting will beat zero-shot, because in-context examples ground the output format and demonstrate what the answer should look like. **H3:** Fine-tuned models will beat prompted models, because EBM-NLP supplies $\sim 4,500$ training documents — well above the scale at which supervised fine-tuning typically dominates few-shot prompting. **H4:** Decomposing extraction into *filter-then-extract* will improve precision, because background and methods sentences that contain no PICO entities are eliminated up-front. **H5:** BioBERT will beat DistilBERT on this corpus, because domain pre-training matches the lexical distribution of biomedical abstracts. Section 3 shows that H1, H3 and H5 hold strongly; H2 fails (zero-shot wins); H4 holds only conditionally on the underlying model.

2.6 Notes on shortcomings of this design

The QA approaches (05_decomposed_qa_roberta.ipynb) target axis B alongside NB06’s decomposed BioBERT pipeline, but their saved predictions were generated against a development-stage document subset before the locked evaluation set was finalised in src/data.py. NB07(notebook 07 for final comparison in notebook folder on git) detects this automatically: any approach with zero coverage on the locked set is flagged and excluded from the master ranking, preventing misleading near-zero F1 values from polluting Table 1. The QA approaches’ internal evaluation in NB05 (macro F1 0.27 end-to-end, 0.39 2-step decomposed, 0.34 3-step) is reported in Table 2 so the decomposed-vs-end-to-end axis remains directly comparable; the underlying QA models are functional, only the doc-ID alignment is stale. NB06’s relevance-classifier + per-field BioBERT

pipeline represents the decomposed extraction axis in the master ranking. **Note:** NB0* refers to respective ipynb file in notebook folder on github

3 Experimental evaluation

3.1 Metrics and their limitations

Token-overlap F1. Treats predictions and gold spans as token sets; $P = |G \cap \hat{Y}|/|\hat{Y}|$, $R = |G \cap \hat{Y}|/|G|$. We follow the convention that both empty $\rightarrow (1, 1, 1)$, gold empty and pred non-empty $\rightarrow (0, 1, 0)$. It is the only definition that applies uniformly across all nine approaches because not all of them emit BIO sequences. *Limitation:* a model returning the single token patients gets partial credit for the gold span “Two hundred and eighty-four out-patients with seasonal allergic rhinitis”. We adopt token-overlap F1 as the report’s **primary metric** because it applies uniformly across all nine approaches; strict span F1 is reported in Table 2 as a stricter secondary check on axis B, where BIO output is available.

Strict span F1. Span match requires identical (start,end,type) in BIO sequences the standard NER metric. Used inside NB06 (CRF, end-to-end BioBERT, decomposed BioBERT) where BIO output is available. *Limitation:* fully penalises minor boundary errors, so a 1-token shift on a 10-token span scores zero.

Coverage. Fraction of documents with at least one non-empty field. A model with high F1 but low coverage is delivering empty rows.

Downstream usability. For each of four keyword queries (*trials with placebo*, *trials mentioning patients*, *trials measuring pain*, *trials with rehabilitation*) we count how many of the 50 extracted PICO rows match the query and compare against the gold count. This is the only metric that is end-user-aligned.

3.2 Setup

All notebooks read EBM-NLP via the shared loader. Training uses the full training split per field (4,454-4,993 docs). Validation for the transformer models is a 10% slice of train; the test split is unused during training. Hyperparameters were chosen to match standard fine-tuning recipes (E2E BioBERT: warm-up by training the head only at 5×10^{-5} before unfreezing at 2×10^{-5}). All runtimes are reported on Google Colab T4 GPU. Predictions for every approach are saved as JSON and aggregated by

Approach	Participants			Interventions			Outcomes			Macro F1
	P	R	F1	P	R	F1	P	R	F1	
E2E BioBERT	.874	.400	.512	.727	.912	.778	.792	.727	.730	.673
DistilBERT	.855	.382	.497	.648	.842	.691	.804	.748	.754	.647
CRF	.609	.229	.301	.540	.624	.514	.692	.510	.563	.459
flan-T5 zero-shot	.843	.301	.392	.696	.423	.434	.493	.154	.215	.347
flan-T5 few-shot	.773	.295	.370	.663	.444	.444	.367	.107	.153	.322
flan-T5 few-shot+JSON	.803	.242	.319	.673	.383	.417	.337	.083	.123	.286
Qwen3-0.6B	.305	.417	.314	.286	.544	.315	.156	.171	.154	.261
Gazetteer	.285	.117	.153	.317	.208	.203	.572	.172	.238	.198
Regex	.313	.299	.262	.126	.079	.083	.344	.066	.106	.150

Table 1: Master comparison on the locked 50-doc test set, ordered by macro token-overlap F1. Best per column **bold**.

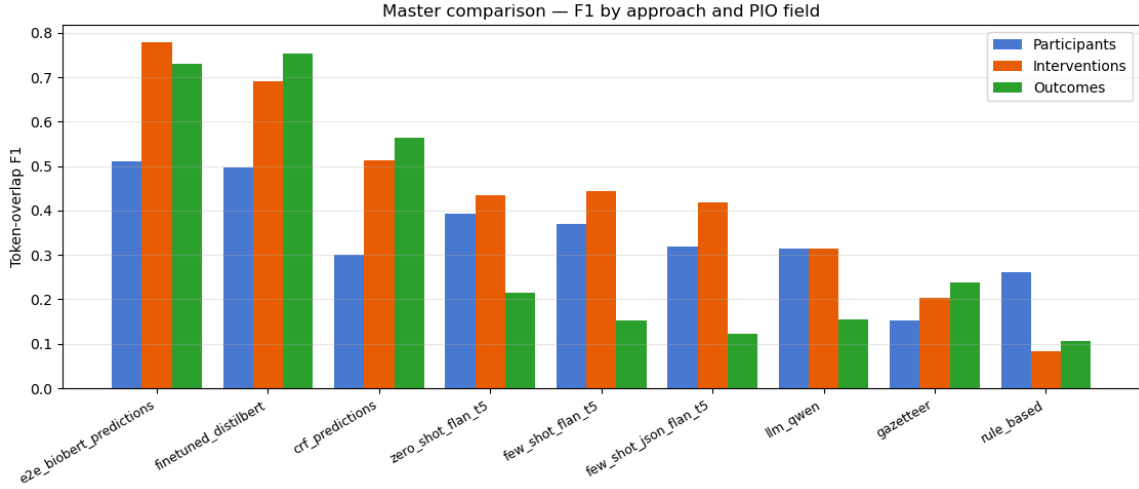


Figure 3: Token-overlap F1 by approach and PICO field on the locked 50-doc test set, ordered by macro F1. Approach labels match Table 1 (e2e_biobert_predictions = E2E BioBERT, finetuned_distilbert = DistilBERT, crf_predictions = CRF, etc.).

NB07; this means re-running NB07 alone reproduces every number in Table 1.

3.3 Master comparison: token-overlap F1

Table 1 reports per-approach, per-field token-overlap P/R/F1 and coverage on the 50-document locked evaluation set, ordered by macro F1. Figure 3 shows the same numbers as a grouped bar chart.

Three observations stand out. First, fine-tuning dominates: the two fine-tuned transformers (BioBERT, DistilBERT) score roughly twice the macro F1 of the best prompted LLM (zero-shot flan-T5, 0.347), confirming H3 strongly — with $\sim 4,500$ training documents available, fine-tuning is the right choice. Second, H1 holds for fine-tuned LLMs but only partially for prompted ones: regex and gazetteer (0.150 and 0.198 macro F1) are beaten by every LLM-based approach including Qwen3-0.6B (0.261), yet the gazetteer beats every

prompted LLM on Outcomes precision (0.572 vs 0.337–0.493) because the highest-frequency outcome 1-grams (score, rate, adverse) are concentrated near the gold spans. Third, Outcomes is the hardest field for every approach: even E2E BioBERT loses almost five points of F1 on Outcomes vs Interventions (0.730 vs 0.778), because outcomes have varied surface forms (*blood pressure*, *quality of life*, *adverse events*) where interventions are usually named drugs or procedures that anchor the answer lexically. We return to this in error analysis.

Statistical caveat on Table 1. With $n = 50$ documents, per-document F1 has substantial variance. The bootstrap analysis we ran on the flan-T5 prompting variants (Table 3) shows 95% CIs typically spanning ± 0.05 F1 at this sample size. Small differences in Table 1 should therefore be interpreted carefully: the gap between E2E BioBERT

Approach	Participants			Interventions			Outcomes			Macro F1
	P	R	F1	P	R	F1	P	R	F1	
<i>NB06: strict span F1 (BIO sequences)</i>										
CRF	.300	.128	.180	.572	.284	.379	.369	.175	.238	.265
E2E BioBERT	.253	.262	.258	.608	.583	.595	.456	.414	.434	.429
Decomposed BioBERT	.225	.252	.237	.605	.366	.456	.400	.376	.388	.360
<i>NB05: token-overlap F1 (extractive QA)</i>										
QA end-to-end	.833	.357	.443	.321	.205	.209	.312	.121	.157	.270
QA decomposed (2-step)	.696	.438	.493	.412	.342	.344	.403	.330	.335	.391
QA decomposed (3-step)	.658	.366	.423	.314	.272	.269	.406	.329	.335	.342

Table 2: Axis B comparisons. NB06 reports strict span F1 (the standard NER metric); NB05 reports token-overlap F1 to be consistent with the master comparison. The two notebooks therefore answer the same question under different difficulty levels.

(0.673) and DistilBERT (0.647) is 0.026 macro F1, well within plausible bootstrap uncertainty, and we treat the top-2 ranking as a tie. Larger gaps e.g. CRF (0.459) versus either fine-tuned transformer, or any fine-tuned model versus any prompted LLM — are well outside this margin and we treat them as reliable.

3.4 Axis B under strict span F1 (secondary metric)

Table 2 reports the within-notebook comparisons for axis B, using each notebook’s own metric so the comparison is internally consistent.

The interaction of the underlying model with the decomposition is the most informative result in the project. **For QA on RoBERTa-SQuAD2**, decomposing helps a lot: 2-step lifts macro F1 from 0.270 to 0.391 (+12 points), driven mainly by Interventions and Outcomes. The relevance step removes background sentences from which the QA model otherwise produces low-confidence spurious spans. **For BioBERT trained directly on token-level annotations**, the opposite happens: decomposing drops macro F1 from 0.429 to 0.360 (−7 points). The TF-IDF + LR classifier filters out around 40% of sentences containing entity tokens (recall drops by 0.05–0.21 across fields while precision is essentially unchanged), and once a sentence is discarded those tokens cannot be recovered. **H4 (decomposing improves precision) therefore holds only when the underlying extractor is the bottleneck;** when the extractor is already strong, decomposing is a net loss.

The 3-step QA pipeline (heuristic cleaning) is uniformly worse than 2-step. The “person word” filter for Participants is too aggressive: a span like “100 elderly Norwegians” is dropped because *Norwegian* is not in the keyword list.

3.5 Statistical reliability of the prompted-LLM comparison

With 50 evaluation documents, per-document F1 is noisy. We ran a paired bootstrap (2,000 resamples) on the three flan-T5 prompting variants (NB03). Mean F1 with 95% confidence intervals is shown in Table 3. Only three pairwise differences are significant at $\alpha = 0.05$: zero-shot beats few-shot+JSON on both Participants and Outcomes, and zero-shot beats few-shot on Outcomes. The remaining six comparisons are within noise.

3.6 Coverage versus correctness, and the placebo failure mode

The four prompted LLMs all report 100% coverage they always emit *something* — but their macro recall sits between 0.24 and 0.42 (Figure 4). The clinical cost of this trade-off is concrete: the gold annotations identify 11 placebo-controlled trials in the locked set, but the four prompted approaches retrieve only 1, 1, 1 and 4 of them respectively when we run the keyword query *trials with placebo* against the extracted Interventions column. Every fine-tuned model and every rule-based approach retrieves at least 11. The gazetteer wins this query outright because placebo is a top-50 training-set term a baseline that costs nothing to build retrieves *more* placebo trials than a 250M-parameter language model. High coverage is not a substitute for high recall on the surface forms that downstream queries depend on.

3.7 Error analysis

We grouped per-document errors by failure mode using NB03’s analysis cells (false positives, false negatives, empty outputs) plus inspection of the worst 10 documents per approach. Five patterns emerged.

Approach	Participants		Interventions		Outcomes	
	F1	95% CI	F1	95% CI	F1	95% CI
flan-T5 zero-shot	.360	[.284, .435]	.385	[.316, .454]	.198	[.149, .250]
flan-T5 few-shot	.339	[.261, .421]	.393	[.332, .452]	.143	[.107, .180]
flan-T5 few-shot+JSON	.289	[.214, .369]	.366	[.300, .431]	.114	[.078, .154]
zero-shot — few-shot	+.020	[−.045, +.086]	−.008	[−.061, +.053]	+.054	[+.004, +.110]
zero-shot — few-shot+JSON	+.070	[+.012, +.134]	+.019	[−.049, +.092]	+.084	[+.037, +.135]
few-shot — few-shot+JSON	+.050	[+.018, +.086]	+.027	[−.018, +.075]	+.030	[−.009, +.065]

Table 3: Per-approach F1 with bootstrap 95% CIs and pairwise differences (paired bootstrap, 2,000 resamples). Boxed values are significant. Note that Interventions is the field where prompt format matters *least* — named drugs anchor the answer regardless.

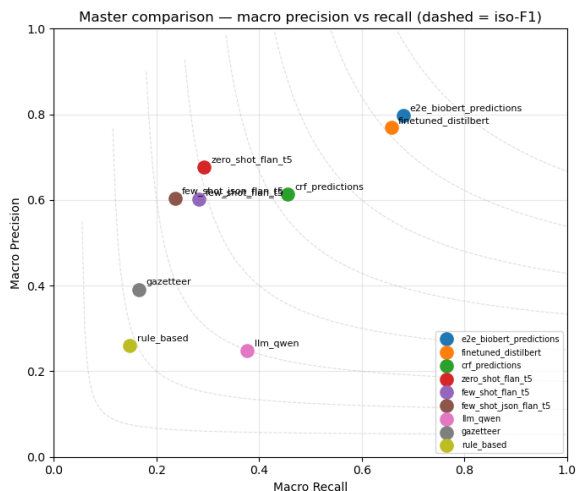


Figure 4: Macro precision vs recall by approach (dashed lines are iso-F1 contours). Prompted LLMs cluster on the high-precision, low-recall side; fine-tuned models reach both.

Few-shot priming bias (prompted LLMs). The few-shot block contains a worked example that mentions “Rhinocort Study Group”. For doc 10070173, the few-shot extractor copies this string verbatim into the Participants field of the new abstract it does not appear in the test abstract at all. Few-shot examples therefore function as soft priors that the model is reluctant to discard. This explains why few-shot loses to zero-shot on Outcomes (where the example shapes the lexical distribution most strongly): the example floods the output with terms that don’t belong to the new trial.

JSON placeholder leakage (NB03 few-shot+JSON). The literal string “not specified” appears in 15 of 50 outcome predictions, by far the most common false-positive token in the JSON condition. The few-shot block contains the fallback “answer”: “not specified” for one

example with no outcome; the model copies it verbatim when uncertain rather than emitting the empty string. The JSON constraint is itself fine; the failure is providing a non-empty placeholder in the example.

Length sensitivity (flan-T5). F1 has a clear negative slope against abstract length for Participants (regression coefficient -2×10^{-3} per token), is approximately flat for Interventions, and trends mildly *positive* for Outcomes. The plausible reason is that flan-T5-base has a 512-token input limit; once the abstract plus three-shot prompt exceeds that limit, the truncation discards the tail of the abstract. Participants are normally introduced in the first 1–2 sentences, so they are robust to truncation only on short abstracts. Outcomes appear throughout, so longer abstracts give the model more correct tokens to land on even after truncation.

Decomposed BioBERT under-recall (NB06). Filter recall on the sentence-classification step is 70–85% per field, but participants sometimes appear in single-sentence abstracts where the methods clause is fused with the patient description (“*Eighty-six male volunteers received a single dose of ...*”). The TF-IDF classifier confuses this with a methods sentence and discards it. Including sentence position as a feature would likely recover most of the lost participants recall.

Outcomes are systematically under-extracted across all approaches. Most-missed gold tokens for E2E BioBERT include *long-term*, *quality-of-life*, *economic* and *ergometric* compound modifiers that the BIO scheme treats as start-of-span on a low-frequency token. Per-field model capacity is not the bottleneck; the annotation scheme is.

Compute cost. Approaches differ by four orders of magnitude in training time. The CRF baseline trains in under 60 seconds on CPU. Fine-tuned DistilBERT trains in roughly 200–600 seconds per field on a Colab T4 GPU, under 30 minutes total across the three fields. End-to-end BioBERT trains in roughly 90 minutes on the same GPU (4 epochs head-only at 5×10^{-5} , then 2 epochs full fine-tune at 2×10^{-5}). The decomposed BioBERT pipeline is the most expensive at ~ 13 hours of CPU-equivalent compute due to per-field training. flan-T5 inference takes 15–25 minutes for 50 documents across three prompting variants on CPU. Whether the 0.21 macro F1 gap from CRF (0.459) to E2E BioBERT (0.673) is worth a $\sim 90\times$ compute increase depends on corpus scale: for a one-off review of $<1,000$ abstracts, CRF is the practical choice; for an evidence-synthesis pipeline running over PubMed, BioBERT amortises over throughput.

3.8 The deliverable: a structured PICO table

The final output of the system is the structured table that the brief asks for. NB07 emits `extracted_pico_best.csv` (best approach, 50 rows, ~ 8 columns) and `extracted_pico_all.csv` (long format with every approach side-by-side). Table 4 shows two example rows from the best approach (E2E BioBERT) for qualitative inspection.

Three observations on the deliverable. First, multi-span fields are preserved by `|`-separated lists, so a downstream query for “trials in children” matches PMID 10390665 even though children is one of three Population spans. Second, the extraction occasionally surfaces noisy inclusion criteria as Participants (e.g. PMID 10589810 reports just `40 | 20`, losing the underlying cohort description) a query-time filter on span length would catch this. Third, comparators remain conflated with interventions (*granisetron* | *perphenazine* for PMID 10390665 mixes the active drug and its comparator), reflecting the EBM-NLP annotation scheme rather than a model failure (§4).

4 Conclusions

Table 5 summarises the verdict on each of the five hypotheses from §2.5 against the experimental data in §3.

H	Claim	Verdict
H1	LLM $>$ rule-based	✓ Confirmed (0.15 macro F1 gap)
H2	Few-shot $>$ zero-shot	× Rejected on Outcomes ($p < 0.05$); n.s. elsewhere
H3	Fine-tuned $>$ prompted	✓ Confirmed strongly (1.94 $\times$ F1)
H4	Decompose $\rightarrow$ better precision	○ Conditional: helps QA, hurts BioBERT
H5	BioBERT $>$ DistilBERT	○ Tie within bootstrap CI (± 0.05)

Table 5: Hypothesis verdicts. ✓ confirmed, × rejected, ○ partial or conditional.

What worked, and what we recommend. For a deployed PICO extraction system over PubMed-scale corpora, we recommend end-to-end BioBERT with the two-phase fine-tuning recipe (head-only at 5×10^{-5} , then full fine-tune at 2×10^{-5}); for a one-off review of fewer than 1,000 abstracts, CRF delivers roughly 70% of the F1 at 1% of the compute and is the better practical choice. Fine-tuning a domain-pretrained transformer on the EBM-NLP training split is the right approach for this task at this data scale: end-to-end BioBERT achieves a macro token-overlap F1 of 0.673, clearly above every other approach we tested.

What did not work as expected. Three of our five hypotheses were challenged by the experiments. Zero-shot beats few-shot for flan-T5 on Outcomes (+0.054 F1, significant) because few-shot examples prime the model with study-specific phrases that leak into unrelated abstracts. Decomposing extraction helps the QA model (+0.12 macro F1) but hurts BioBERT (−0.07 strict span F1), because the TF-IDF relevance filter is the bottleneck of the pipeline rather than the extractor. Adding a JSON output constraint hurts every field, mainly because the literal placeholder string in the few-shot example leaks into 30% of empty cases.

Open challenges. Four challenges remain. First, comparator extraction: EBM-NLP folds the comparator into the Interventions field by design, and although NB05 implements a keyword-based post-processing step (*vs*, *compared to*, *control*, *placebo*) that splits intervention spans, we could not evaluate it quantitatively against any reference annotation; building a re-annotation of EBM-NLP with a separate Comparator field is a clear next step. Second, the coverage–correctness gap: a 100% coverage rate masks the fact that prompted LLMs retrieve 1 out of 11 placebo-controlled trials on a down-

PMID	Participants	Interventions	Outcomes
10390665	children 90 paediatric aged 4–10 years	granisetron perphenazine	Anti-emetic efficacy post-operative complete response
10589810	40 20	root instrumentation conventional steel plastic currettes	bleeding on probing (BOP) probing pocket depth (PPD)

Table 4: Two sample rows from `extracted_pico_best.csv` (E2E BioBERT). The Population is sometimes split into multiple spans (“children” + sample size + age) which the table preserves with | separators; downstream queries can match against any.

stream keyword query, so any deployed system should report a recall-at-query metric in addition to F1 because the latter does not penalise systematic absence of common surface forms. Third, length sensitivity: the 512-token input limit of `flan-T5`-base imposes a hard ceiling on Participants extraction quality for long abstracts, and either a model with a longer context window (e.g. `Long-T5` (Guo et al., 2022)) or a sliding-window inference strategy would address this without changing the prompting axis. Fourth, annotation cost: the compound-modifier failure mode in the error analysis suggests that EBM-NLP’s BIO scheme under-represents multi-word outcome terms; future work should re-annotate at the chunk level for fairer evaluation of long outcome spans.

Limitations of this study. The locked evaluation set is 50 documents chosen so that every notebook can be re-run on a single T4 GPU and bootstrap CIs at this size are typically ± 0.05 F1. Differences smaller than this should be treated as noise. The QA predictions in NB05 were generated against a pre-locked document set and therefore appear with zero coverage on the master comparison; we report them in Table 2 so the decomposed-vs-end-to-end axis remains visible.

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